Applied differential equations – from pendulum to heat conduction

Almost all the laws of physics are differential equations: they say how a state changes from moment to moment and from place to place. Newton's second law, Maxwell's equations, the Schrödinger equation and the heat equation are all of this kind. The course "Applied Differential Equations" was about solving them – and about understanding the solutions even when they cannot be found exactly. The page has an interactive phase plane and a solution of the heat equation with Fourier series.

> University of Sussex, Applied Differential Equations (1975–78): One of the mathematics courses in the physics syllabus. The page describes the subject as it is normally taught in a physics degree; it is not a reproduction of the original course.

First order

The simplest differential equation is $dy/dt = ky$: the change is proportional to the amount itself. The solution $y = y_0 e^{kt}$ describes radioactive decay ($k < 0$), compound interest and bacterial growth ($k > 0$), and the charging of a capacitor. A separable equation $dy/dt = g(t)h(y)$ is solved by gathering y on one side and t on the other and integrating. A linear first-order equation $y' + p(t)y = q(t)$ is solved with an integrating factor $e^{\int p\,dt}$. Newton's law of cooling $dT/dt = -k(T - T_{\text{amb}})$ is an example: the coffee approaches room temperature exponentially.

Oscillations and resonance

Perhaps the most important equation in physics is the damped, driven harmonic oscillator:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2 x = \frac{F_0}{m}\cos\omega t$$

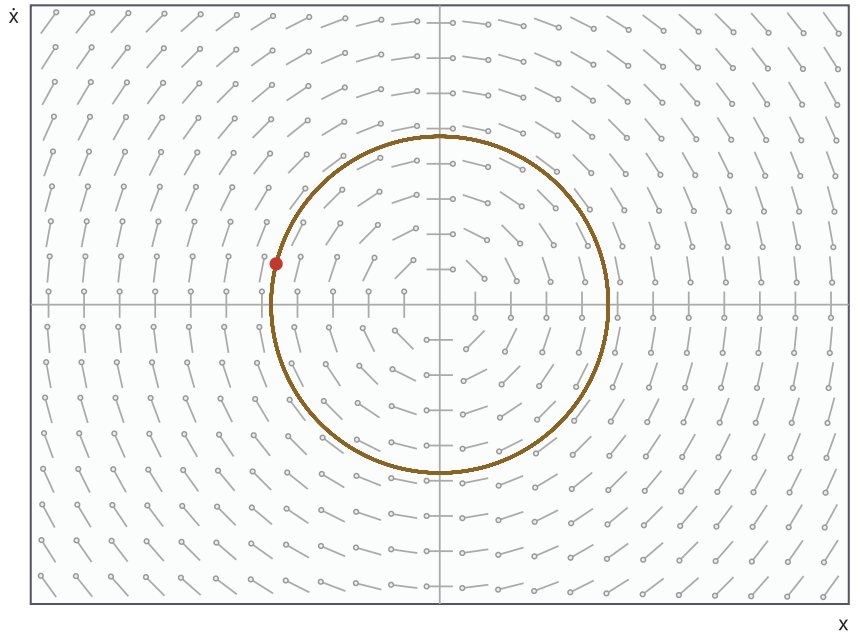
It describes a mass on a spring, an RLC circuit, a loudspeaker cone, an atom in a light field and a building in an earthquake. With the trial solution $x = e^{\lambda t}$ the differential equation becomes a quadratic, the characteristic equation $\lambda^2 + 2\zeta\omega_0\lambda + \omega_0^2 = 0$. Its roots decide whether the system oscillates ($\zeta < 1$), is critically damped ($\zeta = 1$) or creeps ($\zeta > 1$). The general solution is the homogeneous solution plus a particular solution, and after the transient the system oscillates at the driving frequency with the amplitude

$$A(\omega) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\zeta\omega_0\omega)^2}}$$

which has a maximum – resonance – near ω_0. The Tacoma Narrows Bridge, which collapsed in 1940, is often cited as an example, but there it was really an aerodynamic instability (flutter) and not simple resonance.

The phase plane

Henri Poincaré realised around 1880 that a differential equation can be understood without solving it. A system of two first-order equations, $\dot{x} = f(x, y)$, $\dot{y} = g(x, y)$, defines a vector field in the plane, and the solutions are curves that follow the field. For an oscillation x is the position and $y = \dot{x}$ the velocity – a phase plane. Equilibria are points where the field is zero, and their type – node, saddle, spiral or centre – is set by the eigenvalues of the linearised system.



Stability and limit cycles

Linearising about an equilibrium gives a linear system $\dot{\vec{u}} = J\vec{u}$ with the Jacobian matrix J. If all eigenvalues have negative real part, the point is stable; if one has positive real part, it is unstable. Non-linear systems can also have *limit cycles*: isolated closed orbits that neighbouring trajectories approach. Balthasar van der Pol found them in the 1920s in valve oscillators, and they describe the heartbeat, the running of a clock and a violin string vibrating under the bow. With three or more variables the motion can become chaotic – Edward Lorenz discovered this in 1963 in a simplified weather model (see the three-body problem).

Series solutions and special functions

Many equations with variable coefficients can be solved with power series $y = \sum a_n x^n$ (or the Frobenius method $x^r \sum a_n x^n$). That is how the special functions of physics arise: Legendre polynomials from Laplace's equation in spherical coordinates, Bessel functions from cylindrical problems – the vibrations of a drumhead and sound in pipes – Hermite polynomials from the quantum oscillator, and Laguerre polynomials from the hydrogen atom (see quantum mechanics). They are all eigenfunctions of Sturm–Liouville problems and form complete, orthogonal sets, so that an arbitrary function can be expanded in them – just as in a Fourier series.

Partial differential equations

When a quantity depends on both time and position, the equations become partial. The three classic types are

$$\frac{\partial u}{\partial t} = D\nabla^2 u \ \text{(heat)}, \qquad \frac{\partial^2 u}{\partial t^2} = c^2\nabla^2 u \ \text{(waves)}, \qquad \nabla^2 u = 0 \ \text{(Laplace)}$$

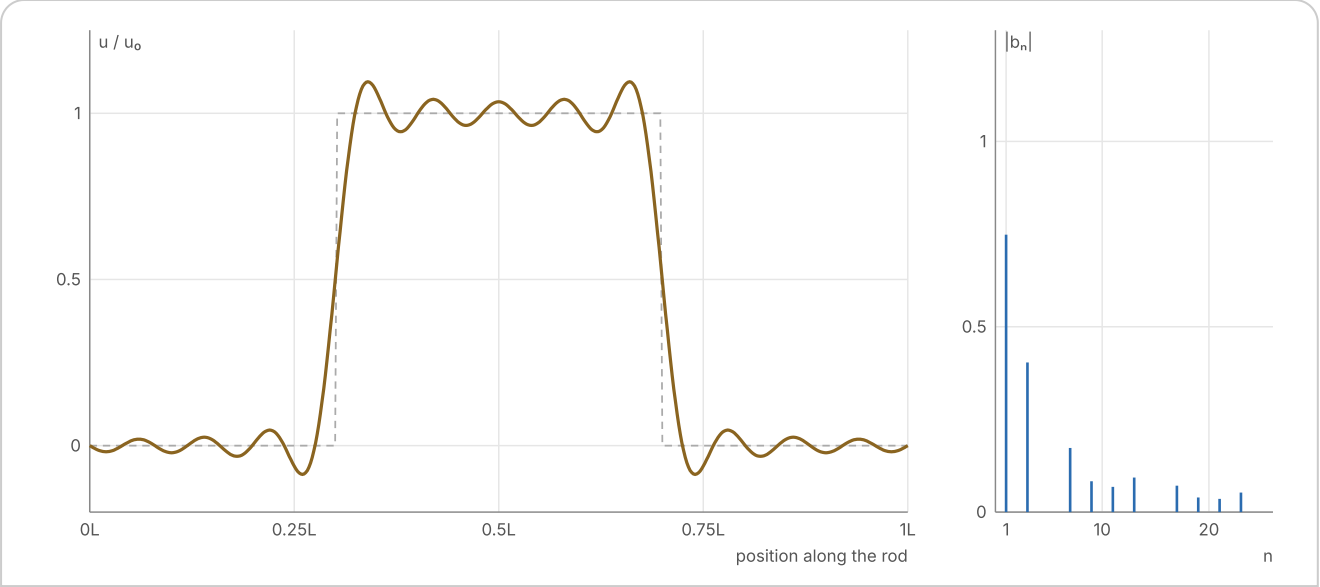
They are parabolic, hyperbolic and elliptic respectively, and they behave fundamentally differently: heat conduction smooths out every detail, waves preserve them and carry them at finite speed, and Laplace solutions are determined by the boundary alone. The classic method is *separation of variables*: one tries a solution of the form $u = X(x)T(t)$, and the partial equation falls apart into ordinary differential equations for each factor.

Heat conduction and Fourier

Joseph Fourier developed his theory precisely to solve the heat equation (1807–1822). For a rod of length L whose ends are held at 0 °C, separation of variables gives the solutions $\sin(n\pi x/L)\, e^{-D(n\pi/L)^2 t}$. The general solution is a sum of them, with coefficients set by the initial state:

$$u(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi x}{L}\, e^{-D(n\pi/L)^2 t}, \qquad b_n = \frac{2}{L}\int_0^L u(x,0)\sin\frac{n\pi x}{L}\,dx$$

Each term dies away at its own rate, proportional to n^2 – the sharp details vanish first, and after a short time only the fundamental remains.



Since 1978

- **Chaos and dynamical systems:** From the mid-1970s (Feigenbaum 1975, Mandelbrot, Ruelle–Takens) chaos theory became a field in its own right; non-linear differential equations are now seen through attractors, bifurcations and fractals.
- **Numerical solvers:** Where exact solutions once had to be found, the computer now solves almost any equation numerically – see numerical mathematics. That has made qualitative methods such as the phase plane more important, not less.
- **Symbolic computation:** Computer algebra (Macsyma 1968, Maple, Mathematica 1988) now solves standard equations and series expansions that once needed tables.
- **New applications:** Epidemic models (SIR) became everyday news during the covid-19 pandemic – a system of three coupled non-linear differential equations.

Further reading

- Steven H. Strogatz: *Nonlinear Dynamics and Chaos* (2nd ed., CRC 2015)
- William E. Boyce and Richard C. DiPrima: *Elementary Differential Equations and Boundary Value Problems* (11th ed., Wiley 2017)

- On this site: Coupled pendulums · Joseph Fourier · The three-body problem